EXERCISE 10

Uniqueness via an Integrating Transformation

Original statement · ProblemsCh2.pdf, p. 3

Linear homogeneous • Uniqueness without Picard

PROBLEM STATEMENT

Without using Picard's theorem, show that the IVP

$$\frac{dy}{dt} + p(t)\, y = 0, \qquad y(t_0) = y_0$$

with $p(t)$ a continuous function has a unique solution, and calculate it. What solution do we obtain if $y_0 = 0$? Hint: consider the function $z(t) = y(t)\, e^{\int_{t_0}^{t} p(s)\, ds}$, where $y(t)$ solves the IVP, show that $z(t)$ is constant and deduce $y(t)$.

Source: ProblemsCh2.pdf, Exercise 10 (page 3).

PHASE 1: CLASSIFICATION, HYPOTHESES AND DOMAIN

Data: p continuous on an open interval $I \ni t_0$; $y_0 \in \mathbb{R}$ arbitrary. A solution is a differentiable function $y : J \to \mathbb{R}$ on an interval $J \subseteq I$ containing t_0 with $y' = -p(t)\, y$ and $y(t_0) = y_0$. Linearity of the equation means that no non-linear terms can cause finite-time blow-up, so we may expect solutions on all of I.

We must prove two statements: (a) any solution is given by one explicit formula (this gives uniqueness), (b) that formula really is a solution (existence). Picard's theorem is not to be used; only the Fundamental Theorem of Calculus and the fact that a function with zero derivative on an interval is constant (Mean Value Theorem) are allowed.

PHASE 2: CHOICE OF METHOD AND CHANGE OF VARIABLES

Why this method (the hint): the function $P(t) = \int_{t_0}^{t} p(s)\,ds$ is differentiable with $P' = p$ because p is continuous (Fundamental Theorem of Calculus). The function $e^{P(t)}$ is exactly an integrating factor μ for $y' + py = 0$. For any solution y, the product $z = y\,e^{P}$ should have zero derivative, which forces z to be constant.

Change of unknown, written explicitly: $z(t) = y(t)\,e^{P(t)}$, $P(t) = \int_{t_0}^{t} p(s)\,ds$, $P(t_0) = 0$, hence $z(t_0) = y_0$.

METHOD OF THE CLASS NOTES [NOTES, P. 3]

The hint is Step 1 of the three-step algorithm of the notes with $q = 0$: $P(t) = \int_{t_0}^{t} p(s)\,ds$, $P(t_0) = 0$, $I(t_0) = 1$, and $z = I\,y$. Step 2 with $q = 0$ is $(Iy)' = 0$, and Step 3 gives $y(t) = \frac{y_0}{I(t)} = y_0\,e^{-P(t)}$. What the notes take for granted (that every solution has this form) is what this problem proves: the zero derivative forces z to be constant by the Mean Value Theorem, with no appeal to Picard.

PHASE 3: STEP-BY-STEP DERIVATION

Uniqueness

Let y be any solution. Differentiate $z = y\,e^{P}$ with the product rule and the chain rule ($\frac{d}{dt}e^{P} = P'\,e^{P} = p\,e^{P}$):

$$z'(t) = y'(t)\,e^{P(t)} + y(t)\,p(t)\,e^{P(t)} = e^{P(t)}\,[y'(t) + p(t)\,y(t)] = e^{P(t)} \cdot 0 = 0.$$

A function whose derivative vanishes on an interval is constant (Mean Value Theorem), so $z(t) = z(t_0)$ for all $t \in J$. Evaluate at t_0: $z(t_0) = y(t_0)\,e^{0} = y_0$. Hence $y(t)\,e^{P(t)} = y_0$, and since $e^{P(t)} > 0$,

$$y(t) = y_0\,e^{-P(t)} = y_0 \exp\left(-\int_{t_0}^{t} p(s)\,ds\right).$$

Every solution coincides with this expression, so there is at most one solution.

Existence

Define $y(t) = y_0\,e^{-P(t)}$ on I. It is differentiable, with $y' = y_0\,e^{-P} \cdot (-P') = -p(t)\,y(t)$, i.e. $y' + py = 0$, and $y(t_0) = y_0 e^{0} = y_0$. So the IVP has this solution on all of I.

The case $y_0 = 0$

The formula gives $y(t) = 0 \cdot e^{-P(t)} \equiv 0$, the trivial solution. By the uniqueness just proved it is the only solution with $y(t_0) = 0$. Equivalently: if a solution vanishes at one time t_1, apply the result with initial time t_1 to get $y \equiv 0$; so a non-trivial solution of this equation never vanishes (since $e^{-P} > 0$, it keeps the sign of y_0).

PHASE 4: VERIFICATION, LIMITS AND INTERPRETATION

Check with $p(t) = 2t, t_0 = 0, y_0 = 3$: $P = t^2, y = 3e^{-t^2}, y' = -6t\, e^{-t^2} = -2t\, y$. Correct.

The same formula follows from the integrating-factor method (Problem 2.3): $\mu = e^{P}$, $(\mu y)' = 0$. The point of this exercise is that the argument above proves uniqueness directly, i.e. it does not presuppose the formula.

Interval of existence: the solution exists on the whole interval I where p is continuous, in agreement with the explicit formula, which is finite wherever p is continuous. This contrasts with non-linear equations such as Problem 2.13, where solutions can blow up in finite time.

RESULT

$$y(t) = y_0 \exp\left(-\int_{t_0}^{t} p(s)\, ds\right), \qquad y_0 = 0 \implies y \equiv 0.$$